Backend Take Home Challenge Description - Checkbox Detection

Backend Challenge: Checkbox Detection

One of the active areas of development at HomeVision is extracting structured information from images. The challenge for you is to develop a system that detects and classifies checkboxes in document images.

The Problem

You are asked to design and implement a system that:

1. **Detects the location** of all checkboxes (filled and unfilled) within a document image.
2. **Classifies each checkbox** as either filled or unfilled.

Perfect accuracy is not expected.

Requirements

Your system must expose an **HTTP API** with the following endpoint:

```
POST /detect
```

Input: A document image (sent as a file upload).

Output: A JSON response with the following structure:

```
{
  "boxes": [
    {
      "bbox": [x1, y1, x2, y2],
```

```
      "is_checked": true
    },
    {
      "bbox": [x1, y1, x2, y2],
      "is_checked": false
    }
  ]
}
```

Where `bbox` contains the pixel coordinates of the top-left and bottom-right corners of each detected checkbox.

Sample Images

Here are 4 document images you can use to test your system against. These images represent real-world appraisal documents with varying characteristics.

Uniform Residential Appraisal Report

File # 000000000

The purpose of this summary appraisal report is to provide the lender/client with an accurate, and adequately supported, opinion of the market value of the subject property.

SUBJECT

Property Address 742 Evergreen Terrace | City Springfield | State IL | Zip Code 62704

Borrower Homer Simpson | Owner of Public Record Homer Simpson | County Sangamon County

Legal Description

Assessor's Parcel # | Tax Year 2022 | R.E. Taxes $ 515

Neighborhood Name | Map Reference Bay | Census Tract 14.02

Occupant [X] Owner [ ] Tenant [ ] Vacant | Special Assessments $ 0 | [ ] PUD | HOA $ 0 | [ ] per year [ ] per month

Property Rights Appraised [X] Fee Simple [ ] Leasehold [ ] Other (describe)

Assignment Type [ ] Purchase Transaction [X] Refinance Transaction [ ] Other (describe)

Lender/Client Loan Services, LLC | Address

Is the subject property currently offered for sale or has it been offered for sale in the twelve months prior to the effective date of the appraisal? [ ] Yes [X] No

Report data source(s) used, offering price(s), and date(s). DOM 0;

CONTRACT

I [ ] did [ ] did not analyze the contract for sale for the subject purchase transaction. Explain the results of the analysis of the contract for sale or why the analysis was not performed.

Contract Price$ | Date of Contract | Is the property seller the owner of public record? [ ] Yes [ ] No | Data Source(s)

Is there any financial assistance (loan charges, sale concessions, gift or downpayment assistance, etc.) to be paid by any party on behalf of the borrower? [ ] Yes [ ] No

If Yes, report the total dollar amount and describe the items to be paid:

NEIGHBORHOOD

Note: Race and the racial composition of the neighborhood are not appraisal factors.

Neighborhood Characteristics			One-Unit Housing Trends			One-Unit Housing			Percent Land Use %	
Location	[X] Urban	[ ] Suburban [ ] Rural	Property Values	[ ] Increasing	[X] Stable [ ] Declining	PRICE	AGE		One-Unit	93 %
Built-Up	[X] Over 75%	[ ] 25-75% [ ] Under 25%	Demand/Supply	[ ] Shortage	[X] In Balance [ ] Over Supply	$(000)	(yrs)		2-4 Unit	1 %
Growth	[ ] Rapid	[X] Stable [ ] Slow	Marketing Time	[X] Under 3 mths	[ ] 3-6 mths [ ] Over 6 mths	85	Low	0	Multi-Family	0 %
Neighborhood Boundaries	north by North Bay east by Cato Rd south by 26th St west by Hwy 77.					1,700	High	110	Commercial	6 %
						361	Pred.	30	Other	0 %

Neighborhood Description The subjects neighborhood has typical residential amenities which include schools, shopping and medical facilities.

Market Conditions (including support for the above conclusions) The current market conditions are stable with typical marketing times from 0-3 months.

SITE

Dimensions 100Fx150LSx100Rx150RS | Area 15000 sf | Shape rectangular | View N;Res;

Specific Zoning Classification RLD | Zoning Description single family

Zoning Compliance [X] Legal [ ] Legal Nonconforming (Grandfathered Use) [ ] No Zoning [ ] Illegal (describe)

Is the highest and best use of the subject property as improved (or as proposed per plans and specifications) the present use? [X] Yes [ ] No If No, describe

Utilities	Public	Other (describe)	Public	Other (describe)	Off-site Improvements--Type	Public	Private
Electricity	[X]	[ ]	Water	[X] [ ]	Street asphalt	[X]	[ ]
Gas	[ ]	[ ] none	Sanitary Sewer	[X] [ ]	Alley none	[ ]	[ ]

FEMA Special Flood Hazard Area [ ] Yes [X] No | FEMA Flood Zone X | FEMA Map No. | FEMA Map Date 06/02/2009

Are the utilities and off-site improvements typical for the market area? [X] Yes [ ] No. If No, describe

Are there any adverse site conditions or external factors (easements, encroachments, environmental conditions, land uses, etc.)? [ ] Yes [X] No If Yes, describe

IMPROVEMENTS

General Description		Foundation		Exterior Description materials/condition		Interior materials/condition	
Units	[X] One [ ] One with Accessory Unit	[X] Concrete Slab	[ ] Crawl Space	Foundation Walls	concrete/Avg	Floors	tile,wood lam/Avg
# of Stories	1	[ ] Full Basement	[ ] Partial Basement	Exterior Walls	stucco/Avg	Walls	Sheetrock/Avg
Type	[X] Det. [ ] Att. [ ] S-Det/End Unit	Basement Area	0 sq. ft.	Roof Surface	comp shingle/Gd	Trim/Finish	wood/Avg
[X] Existing	[ ] Proposed [ ] Under Const.	Basement Finish	0 %	Gutters & Downspouts	none	Bath Floor	tile/Avg
Design (Style)	rambler	[X] Outside Entry/Exit	[ ] Sump Pump	Window Type	vinyl/Gd	Bath Wainscot	tile/Avg
Year Built	1950	Evidence of	[ ] Infestation	Storm Sash/Insulated	none	Car Storage	[ ] None
Effective Age (Yrs)	20	[ ] Dampness	[ ] Settlement	Screens	nylon/Gd	[X] Driveway	# of Cars 2
Attic	[ ] None	Heating [X] FWA	[ ] HWBB [ ] Radiant	Amenities	[ ] Woodstove(s) # 0	Driveway Surface	concrete
[X] Drop Stair	[ ] Stairs	[ ] Other	Fuel electric	[X] Fireplace(s) # 1	[X] Fence wood	[ ] Garage	# of Cars 0
[ ] Floor	[ ] Scuttle	Cooling [X] Central Air Conditioning		[X] Patio/Deck crvd	[X] Porch crvd	[ ] Carport	# of Cars 0
[ ] Finished	[ ] Heated	[ ] Individual	[ ] Other	[ ] Pool none	[ ] Other none	[ ] Att. [ ] Det.	[ ] Built-in

Appliances [X] Refrigerator [X] Range/Oven [ ] Dishwasher [ ] Disposal [ ] Microwave [X] Washer/Dryer [ ] Other (describe)

Finished area above grade contains: 5 Rooms | 4 Bedrooms | 1.0 Bath(s) | 1,526 Square Feet of Gross Living Area Above Grade

Additional features (special energy efficient items, etc.) windows

Describe the condition of the property (including needed repairs, deterioration, renovations, remodeling, etc.). C4;No updates in the prior 15 years;The subject had a new roof added in 2020, new windows have been added in the last few years.

Are there any physical deficiencies or adverse conditions that affect the livability, soundness, or structural integrity of the property? [ ] Yes [X] No If Yes, describe

Does the property generally conform to the neighborhood (functional utility, style, condition, use, construction, etc.)? [X] Yes [ ] No If No, describe

NEIGHBORHOOD

Note: Race and the racial composition of the neighborhood are not appraisal factors.

Neighborhood Characteristics				One-Unit Housing Trends				One-Unit Housing			Present Land Use %	
Location	☐ Urban	☒ Suburban	☐ Rural	Property Values	☒ Increasing	☐ Stable	☐ Declining	PRICE		AGE	One-Unit	55 %
Built-Up	☐ Over 75%	☒ 25-75%	☐ Under 25%	Demand/Supply	☒ Shortage	☐ In Balance	☐ Over Supply	$ (000)		(yrs)	2-4 Unit	0 %
Growth	☐ Rapid	☒ Stable	☐ Slow	Marketing Time	☒ Under 3 mths	☐ 3-6 mths	☐ Over 6 mths	319	Low	18	Multi-Family	10 %
Neighborhood Boundaries								737	High	50	Commercial	10 %
								525	Pred.	30	Other	5 %

Neighborhood Description: The direct market area of the city limits of Fort Worth is primarily comprised of residential land use of varying chronological age, quality and condition of homes. The area is convenient to major transportation, schools, and recreational facilities. The immediate neighborhood Fort Worth Springs is made up of similar average to good quality single family site built homes with 10,000sf minimum lot size

Market Conditions (including support for the above conclusions): 12 month market data for comparable properties within the City Limits of Fort Worth resulted in 19 properties including 1 active listing and 18 closed sales, with an average sales price of $371,000 and average list price of $375,000. The overall trendline of this data shows appreciation of approximately 8-10% annualized.

SITE

Dimensions: See Plat Map | Area: 10542 sf | Shape: Triangular | View: N;Res;

Specific Zoning Classification: Rs100 | Zoning Description: Residential Single Family (Min. Property: 10,000 sqft.)

Zoning Compliance: ☒ Legal ☐ Legal Nonconforming (Grandfathered Use) ☐ No Zoning ☐ Illegal (describe)

Is the highest and best use of subject property as improved (or as proposed per plans and specifications) the present use? ☒ Yes ☐ No If No, describe

Utilities	Public	Other (describe)		Public	Other (describe)	Off-site Improvements - Type	Public	Private
Electricity	☒	☑	Water	☒	☐	Street: Asphalt	☒	☐
Gas	☒	☐	Sanitary Sewer	☒	☐	Alley: None	☐	☐

FEMA Special Flood Hazard Area ☐ Yes ☒ No | FEMA Flood Zone: X | FEMA Map #: | FEMA Map Date: 09/25/2009

Are the utilities and off-site improvements typical for the market area? ☒ Yes ☐ No If No, describe

Are there any adverse site conditions or external factors (easements, encroachments, environmental conditions, land uses, etc.)? ☐ Yes ☒ No If Yes, describe

Appraisal Group, LLC

Market Conditions Addendum to the Appraisal Report

File No.
Case No. 000000000

The purpose of this addendum is to provide the lender/client with a clear and accurate understanding of the market trends and conditions prevalent in the subject neighborhood. This is a required addendum for all appraisal reports with an effective date on or after April 1, 2009.

Property Address 4, Private Drive | City Little Whinging | State X | ZIP Code

Borrower Vernon Dursley

Instructions: The appraiser must use the information required on this form as the basis for his/her conclusions and must provide support for those conclusions, regarding housing trends and overall market conditions as reported in the Neighborhood section of the appraisal report form. The appraiser must fill in all the information to the extent it is available and reliable and must provide analysis as indicated below. If any required data is unavailable or is considered unreliable, the appraiser must provide an explanation. It is recognized that not all data sources will be able to provide data for the shaded areas below; if it is available, however, the appraiser must include that data in the analysis. If data sources provide all the required information as an average instead of the median, the appraiser should report the available figure and identify it as an average. Sales and listings must be properties that compete with the subject property, determined by applying the criteria that would be used by a prospective buyer of the subject property. The appraiser must explain any anomalies in the data, such as seasonal markets, new construction, foreclosures, etc.

MARKET RESEARCH & ANALYSIS

Inventory Analysis	Prior 7-12 Months	Prior 4-6 Months	Current - 3 Months	Overall Trend
Total # of Comparable Sales (Settled)	28	7	9	[X] Increasing [ ] Stable [ ] Declining
Absorption Rate (Total Sales/Months)	4.67	2.33	5.67	[X] Increasing [ ] Stable [ ] Declining
Total # of Comparable Active Listings	13	9	8	[X] Declining [ ] Stable [ ] Increasing
Months of Housing Supply (Total Listings/Ab. Rate)	2.78	3.86	1.41	[X] Declining [ ] Stable [ ] Increasing
Median Sales & List Price, DOM, Sale/List %	Prior 7-12 Months	Prior 4-6 Months	Current - 3 Months	Overall Trend
Median Comparable Sales Price	279,950.00	275,000.00	283,900.00	[ ] Increasing [X] Stable [ ] Declining
Median Comparable Sales Days on Market	6	7	132	[ ] Declining [ ] Stable [X] Increasing
Median Comparable List Price	359,900.00	359,900.00	359,900.00	[ ] Increasing [X] Stable [ ] Declining
Median Comparable Listings Days on Market	161	149	14	[X] Declining [ ] Stable [ ] Increasing
Median Sale Price as % of List Price	100.00	104.00	100.00	[ ] Increasing [ ] Stable [X] Declining
Seller-(developer, builder, etc.) paid financial assistance prevalent?	[X] Yes [ ] No			[X] Declining [ ] Stable [ ] Increasing

Explain in detail seller concessions trends for the past 12 months (e.g. seller contributions increased from 3% to 5%, increasing use of buydowns, closing costs condo fees, options, etc.)

Concessions reported have been decreasing to 0.82% from 1.09% 12 months back. Since real estate agents are not required to report concessions by Georgia MLS, only reported concessions were analyzed.

Are foreclosure sales (REO sales) a factor in the market? [X] Yes [ ] No If yes, explain (including the trends in listings and sales of foreclosed properties).

During the past year, a significant proportion of sold listings and active listing have been bank-owned properties (REO sales). The excessive number of REO's have contributed to the decline in market value over the past year. However, the data for the area indicates a trend towards a stabilization over the previous six months.

Cite data sources for above information.

MLS - The information was derived from a 1 mile radius search around the subject property.

Summarize the above information as support for your conclusions in the Neighborhood section of the appraisal report form. If you used any additional information, such as an analysis of pending sales, and/or expired and withdrawn listings, to formulate your conclusions, provide both an explanation and support for your conclusions.

There were a total of 44 Comparable Settled Sales in the past 12 months.
The Median Sales Price for the prior 7-12 months was $279,950 the current to prior 3 months is $349,900.
Months Supply for the prior 7-12 months was 2.8 and 1.4 for the current to prior 3 month period.
The Median Days on Market for the prior 7-12 months was 6 and 132 for the current to prior 13 month period.
The statistics above were generated from an exported MLS market search. Details regarding the calculations and process can be found online at http://bradfordsoftware.com/1004mc/calc.shtml.

CONDO/CO-OP PROJECTS

If the subject is a unit in a condominium or cooperative project, complete the following: Project Name:

Subject Project Data	Prior 7-12 Months	Prior 4-6 Months	Current - 3 Months	Overall Trend
Total # of Comparable Sales (Settled)				[ ] Increasing [ ] Stable [ ] Declining
Absorption Rate (Total Sales/Months)				[ ] Increasing [ ] Stable [ ] Declining
Total # of Active Comparable Listings				[ ] Declining [ ] Stable [ ] Increasing
Months of Unit Supply (Total Listings/Ab. Rate)				[ ] Declining [ ] Stable [ ] Increasing

Are foreclosures sales (REO sales) a factor in the project? [ ] Yes [ ] No If yes, indicate the number of REO listings and explain the trends in listings and sales of foreclosed properties.

Summarize the above trends and address the impact on the subject unit and project.

APPRAISER

Signature

Supervisor Name
Company Name
Company Address
State License/Certification # State
Email Address

Valuation Now Appraisals Inc.

Manufactured Home Appraisal Report

File #

The purpose of this summary appraisal report is to provide the lender/client with an accurate, and adequately supported, opinion of the market value of the subject property.

SUBJECT

Property Address | City | State CA | Zip Code

Borrower | Owner of Public Record | County

Legal Description

Assessor's Parcel # | Tax Year 2024 | R.E. Taxes $ '23

Neighborhood Name | Map Reference 140 | Census Tract 44.06

Occupant [X] Owner [ ] Tenant [ ] Vacant | Project Type (if applicable) [ ] PUD [ ] Condominium [ ] Cooperative [ ] Other (describe)

Special Assessments $ 0 | HOA $ 0 [ ] per year [ ] per month

Property Rights Appraised [X] Fee Simple [ ] Leasehold [ ] Other (describe)

Assignment Type [ ] Purchase Transaction [ ] Refinance Transaction [X] Other (describe)

Lender/Client Financial Management, LLC | Address

Is the subject property currently offered for sale or has it been offered for sale in the twelve months prior to the effective date of this appraisal? [ ] Yes [X] No

Report data source(s) used, offering price(s), and date(s). The subject property is not currently listed nor has been in the past 1 year/12 months per the local CRMLS.

Manufactured homes located in either a condominium or cooperative project require the appraiser to inspect the project and complete the Project Information section of the Individual Condominium Unit Appraisal Report or the Individual Cooperative Interest Appraisal Report and attach it as an addendum to this report.

CONTRACT

I [ ] did [ ] did not analyze the contract for sale for the subject purchase transaction. Explain the results of the analysis of the contract for sale or why the analysis was not performed.

Contract Price $ | Date of Contract | Is the property seller the owner of public record? [ ] Yes [ ] No | Data Source(s)

Is there any financial assistance (loan charges, sale concessions, gift or downpayment assistance, etc.) to be paid by any party on behalf of the borrower? [ ] Yes [ ] No

If Yes, report the total dollar amount and describe the items to be paid.

I [ ] did [X] did not analyze the manufacturer's invoice. Explain the results of the analysis of the manufacturer's invoice or why the analysis was not performed.

No invoice was available.

Retailer's Name (New Construction) N/A

NEIGHBORHOOD

Note: Race and the racial composition of the neighborhood are not appraisal factors.

Neighborhood Characteristics				Manufactured Housing Trends				Manufactured Housing		Present Land Use %	
Location	[ ] Urban	[X] Suburban	[ ] Rural	Property Values	[ ] Increasing	[X] Stable	[ ] Declining	PRICE	AGE	One-Unit	65 %
Built-Up	[ ] Over 75%	[X] 25-75%	[ ] Under 25%	Demand/Supply	[ ] Shortage	[X] In Balance	[ ] Over Supply	$ (000)	(yrs)	2-4 Unit	0 %
Growth	[ ] Rapid	[X] Stable	[ ] Slow	Marketing Time	[X] Under 3 mths	[ ] 3-6 mths	[ ] Over 6 mths	95 Low	1	Multi-Family	0 %
Neighborhood Boundaries	The Area Limits to the North, The area of to the South and the area known as							1,055 High	60	Commercial	5 %
								440 Pred.	30	Other	30 %

Neighborhood Description The subjects neighborhood is located in the unincorporated area known as Estates.This community predominantly consists of one to two-story single-family residences and manufactured homes. It is located within ten miles from schools, shopping centers, and employment centers. 'Other' above under Present Land Use % is open/vacant land.

Market Conditions (including support for the above conclusions) Market conditions have been stable in the subjects market area. There is a steady demand for housing, economic stability, and steady employment. Exposure time is typically 1 to 90 days. Marketing time is 1 to 90 days and based on statistical information for the surrounding area.

SITE

Dimensions 353 X 326 X 354 X 326 ft | Area 2.65 ac | Shape Square | View N;Res;

Specific Zoning Classification R-1-2 1/2 | Zoning Description Residential

Zoning Compliance [X] Legal [ ] Legal Nonconforming (Grandfathered Use) [ ] No Zoning [ ] Illegal (describe)

Is the highest and best use of subject property as improved (or as proposed per plans and specifications) the present use? [X] Yes [ ] No If No, describe The highest and best use of this property as improved, or as vacant, is the existing use. The current use gives the greatest net return to the land.

Utilities	Public	Other (describe)		Public	Other (describe)	Off-site Improvements - Type	Public	Private
Electricity	[X]	[ ]	Water	[ ]	[X] Well	Street Gravel/Dirt	[ ]	[X]
Gas	[ ]	[X] Propane	Sanitary Sewer	[ ]	[X] Septic	Alley None	[ ]	[ ]

FEMA Special Flood Hazard Area [ ] Yes [X] No | FEMA Flood Zone D | FEMA Map # | FEMA Map Date

Are the utilities and off-site improvements typical for the market area? [X] Yes [ ] No If No, describe

Is the site size, shape and topography generally conforming to and acceptable in the market area? [X] Yes [ ] No If No, explain

Is there adequate vehicular access to the subject property? [X] Yes [ ] No If No, describe

Is the street properly maintained? [X] Yes [ ] No If No, describe The streets with in Lake Riverside Estates are maintained by the HOA.

Are there any adverse site conditions or external factors (easements, encroachments, environmental conditions, land uses, etc.)? [ ] Yes [X] No If Yes, describe

The site is located on a residential street surrounded by similarly improved properties. No adverse easements/encroachments noted.

HUD DATA PLATE

The HUD Data Plate/Compliance Certificate is located on the interior of the subject and contains, among other things, the manufacturer's name, trade/model name, year manufactured and serial number. The HUD Certification Label is located on the exterior of each section of the home.

Is the HUD Data Plate/Compliance Certificate attached to the dwelling? [X] Yes [ ] No If Yes, identify the location. If No, provide the data source(s) for the HUD Data Plate/Compliance Certificate information. Under the Kitchen Sink

Is a HUD Certification Label attached to the exterior of each section of the dwelling? [X] Yes [ ] No If No, provide the data source(s) for the HUD Certification Label #'s

The subject has 2 sections, both of which had an exterior HUD label

Date of Manufacture

Do the Wind, Roof Load, and Thermal Zones meet the minimum HUD requirements for the location of the subject property? [X] Yes [ ] No If No, explain

According to the Compliance Certification.

Created with a trial version of Snagit. Buy it at techsmith.com/purchase.html

Guidelines

- You can use **any method and any tools** you want.
- Your solution must include instructions on **how to build and run** it.
- Include a brief **writeup** explaining your approach, the tradeoffs you made, and any known limitations.
- Feel free to use `TODO` comments to identify parts of the solution where you would do additional work for production-quality code.
- We value **your reasoning and decision-making process** as much as the final result. Be prepared to explain why you chose your approach over alternatives.

Submission

When your solution is complete, use the form below to submit a zip of your project files (including your writeup). We will review your solution and get back to you to schedule a one-hour challenge review meeting. In that meeting, you will walk us through your code and approach.

Good luck and have fun!